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(54) **ANTI-FGFR2 ADCC ENHANCED ANTIBODY
AND USE THEREOF**

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(57) **ABSTRACT**

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Provided in the present invention is an anti-FGFR2 ADCC enhanced antibody. The antibody achieves ADCC enhancement by means of enhancing the binding to FcγR. In tumor therapy, the antibody has anti-tumor activity.

Specification includes a Sequence Listing.

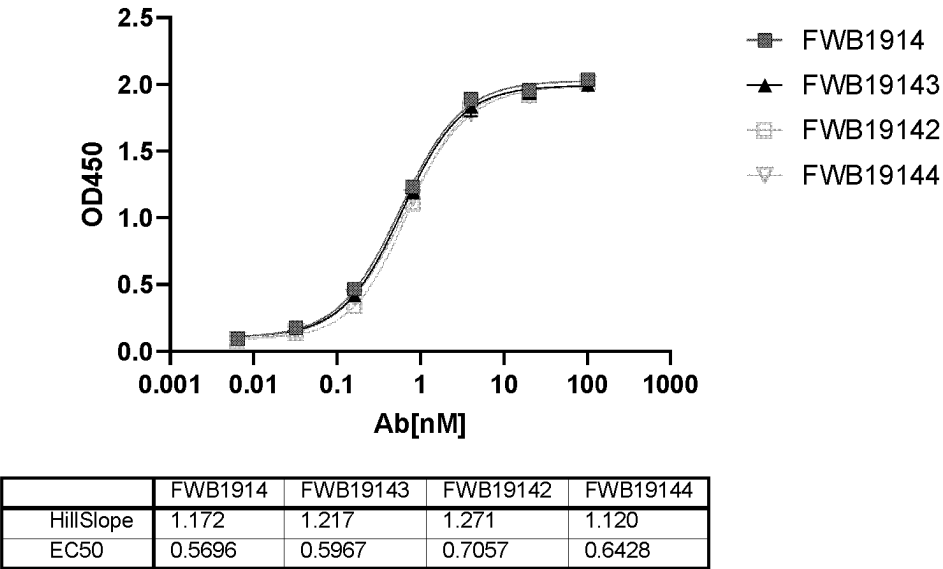
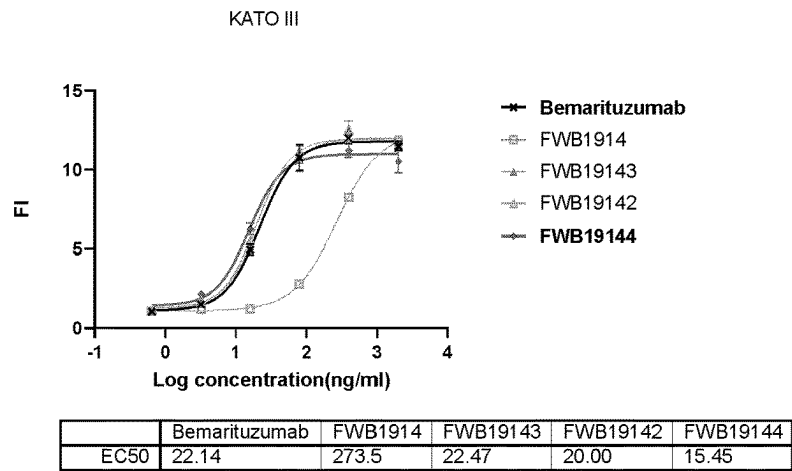
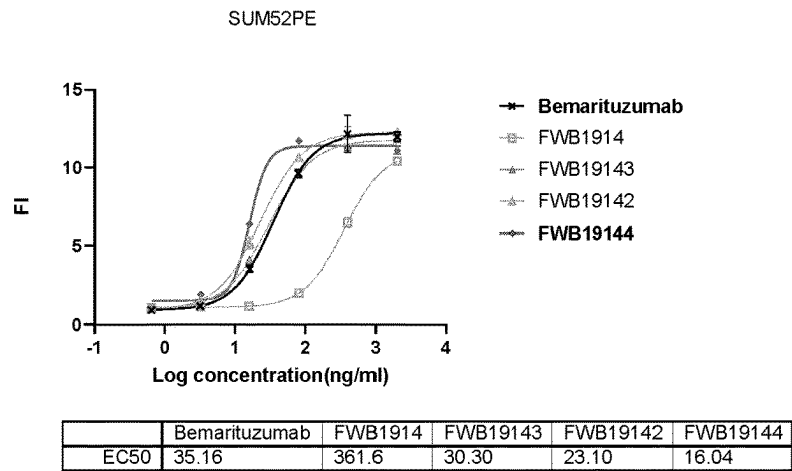


图 1

A



B



C

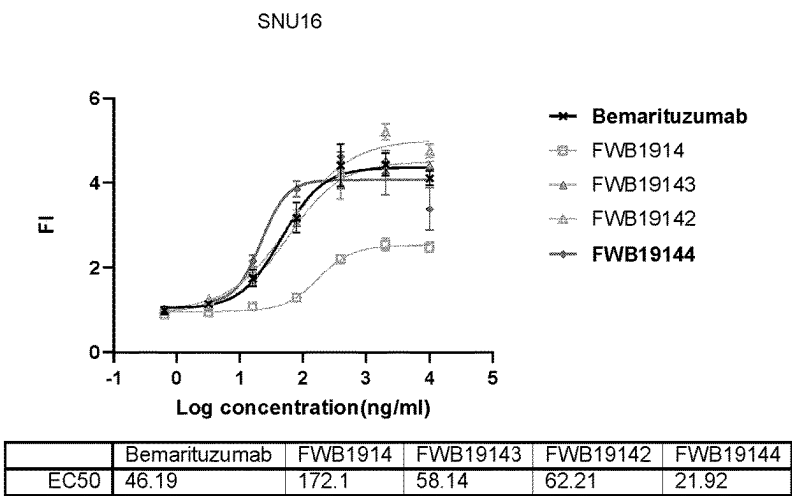
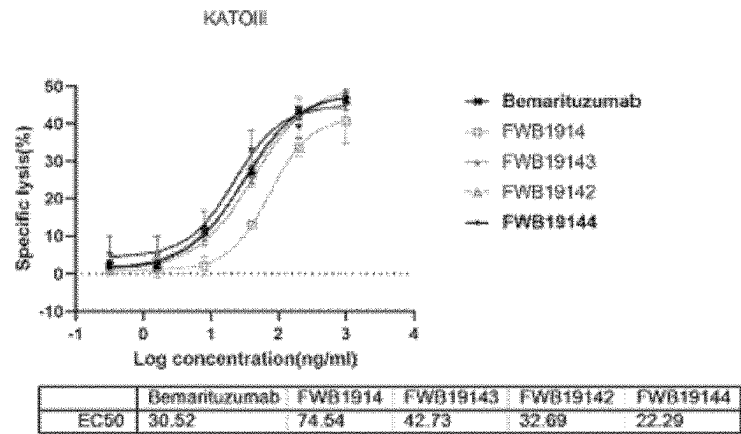
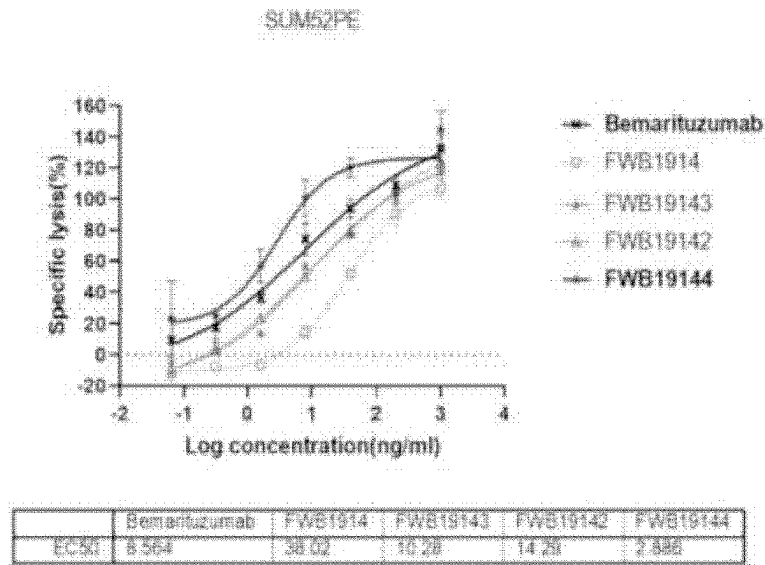


图 2

A



B



C

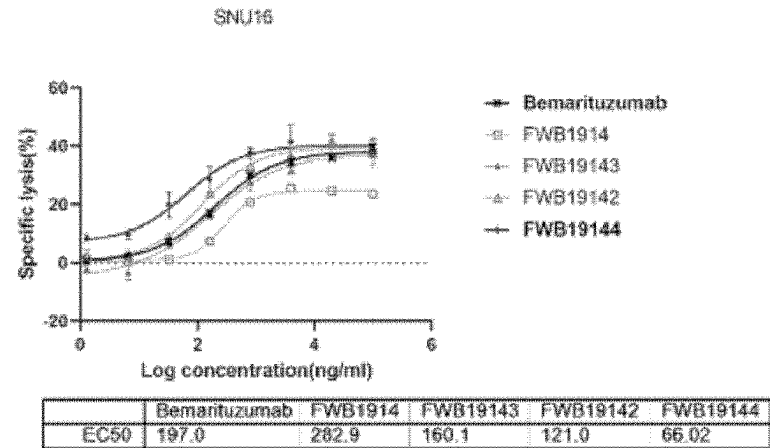


图 3

ANTI-FGFR2 ADCC ENHANCED ANTIBODY AND USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a 35 U.S.C. § 371 national phase application of, and claims priority to, PCT International Application No. PCT/CN2022/123723, filed Oct. 7, 2022, which claims priority to PCT International Application No. PCT/CN2021/122699, filed Oct. 8, 2021, the entireties of which are hereby incorporated herein by reference.

SEQUENCE LISTING

[0002] This application contains a sequence listing in .xml format entitled “F22W0970US-sequence listing.xml” created Sep. 22, 2024 and having a size of 19 KB. The sequence listing contained in this .xml file is part of the specification and is herein incorporated by reference in its entirety.

TECHNICAL FIELD

[0003] The present invention relates to an ADCC-enhanced anti-FGFR2 antibody and use thereof.

BACKGROUND ART

[0004] In normal cells, fibroblast growth factor (FGF) receptor 2 (FGFR2) can transduce FGF signals to the RAS-ERK and PI3K-AKT signaling pathways through the FRS2 cascade, and also regulate cell differentiation, proliferation, and apoptosis by activating the DAG-PKC and IP3-calmodulin signaling cascades with PLC γ .

[0005] The coding gene for FGFR2 is distributed on human chromosome 10q26, and due to splicing changes, FGFR2b and FGFR2c isoforms are produced. The immunoglobulin-like domains and intracellular tyrosine kinase domains of FGFR2b and FGFR2c are almost identical, with the main difference being in the latter half of the third immunoglobulin-like domain. In addition, the main expressing cells and binding ligands of the two isoforms are also different, wherein FGFR2b, which is mainly expressed in epithelial cells, binds strongly to ligands FGF1, FGF3, FGF7, FGF10, and FGF22, and FGFR2c, which is expressed in mesenchymal cells, mainly binds to ligands FGF1, FGF2, FGF4, FGF6, FGF9, FGF16, and FGF20.

[0006] Overexpression of FGFR2 or FGFs, as well as genetic changes such as gene amplification, gene fusion and rearrangement, gene point mutations, and chromosomal translocation, can lead to dysregulation of the FGFR2 signaling pathway, and the dysregulated FGFRs/FGFs signaling pathway is closely related to cellular carcinogenesis. Potential overexpression, activation by missense mutation, or abnormal protein fusion of FGFR2 have been reported in a variety of cancer types, including endometrial cancer, ovarian cancer, breast cancer, lung cancer, gastric cancer, and bile duct cancer. Therefore, FGFR2 can become a potential target for tumor treatment.

[0007] Currently, highly specific anti-tumor drugs targeting FGFR2 are undergoing preclinical and clinical trials. These drugs are classified according to their mechanisms of action as receptor tyrosine kinase inhibitors, antagonistic antibodies, or peptide inhibitors. Among them, several monoclonal antibodies targeting FGFR2 are in the preclinical or clinical development stage. For example, the

FGFR2IIIb-specific antibody GP369 has been shown to inhibit the proliferation of human cancer cell lines and tumor xenografts by amplified or activated FGFR2 signals. BAY1187982 (Bayer) had been shown to be effective in preclinical research by adopting the therapeutic strategy of antibody-drug conjugates. It successfully inhibited tumor growth in the xenotransplantation models of gastric cancer and breast cancer with overexpression of FGFR2. Unfortunately, the Phase I clinical trial of the drug was terminated in 2016 due to safety concerns. The antibody FP144 (Five Prime) targeting FGFR2IIIb has been shown to inhibit FGF/FGFR2-related tumor signaling pathways, thereby inhibiting tumor growth. Phase II clinical trial data shows that in gastric cancer with overexpression of FGFR2, the combination therapy of FP144 and chemotherapeutic drugs demonstrates positive therapeutic effects.

[0008] To improve the therapeutic effects of antibody drugs, many pharmaceutical companies and research institutions have made various attempts. Among them, antibody Fc modification has been proven to be an effective approach to enhance antibody-dependent cell-mediated cytotoxicity (ADCC), thereby enhancing anti-tumor activity. Fucose knockout at Asn297 or Fc point mutations are mainly used in Fc modification for enhancing ADCC.

[0009] IgG1 subtype antibodies are widely used in cancer immunotherapy, and there is a conserved N-linked glycosylation site at Asn297 of the CH2 domain of IgG1 subtype antibodies. Studies have shown that removing the N-linked glycan at this site can enhance binding of the antibody Fc end to Fc γ Rs on NK cells, thereby enhancing ADCC activity. In 1999, Umana, P. et al. reported that the glycosylation content at Asn297 of the antibody produced using a cell line, in which the β (1,4)-N-acetylglucosaminyltransferase III was knocked out in Chinese hamster ovarian cells, decreased, and the antibody exhibited significantly enhanced ADCC activity in vitro. So far, multiple afucosylated antibody drugs have been approved for marketing or are in the clinical stage. For example, Obinutuzumab targeting CD20 developed by Roche and Mogamulizumab targeting CCR4 developed by Kyowa Hakko Kirin have been approved for marketing in the United States and Europe.

[0010] In addition to knocking out fucose, another proven effective strategy is to introduce a point mutation at the Fc end, in order to develop mutants with high affinity for Fc γ Rs. Based on in silicon interaction analysis, Xencor found that when S239D/1332E or S239D/1332E/A330L were introduced into IgG1 Fc, the results showed that compared to the wild-type, the mutant exhibited an affinity for Fc γ Rs increased by 100 times, and significantly enhanced ADCC activity.

SUMMARY OF THE INVENTION

[0011] The present invention provides an ADCC-enhanced anti-FGFR2 antibody and use thereof. In one embodiment, ADCC is enhanced by enhancing binding to Fc γ Rs.

[0012] In one aspect, the present invention provides an ADCC-enhanced anti-FGFR2 antibody, wherein compared to a wild-type Fc region that is fucosylated, the Fc region of the antibody exhibits enhanced binding to Fc γ Rs.

[0013] In one embodiment, the heavy chain variable region of the antibody comprises an HVR-H1 as set forth in SEQ ID NO: 5, an HVR-H2 as set forth in SEQ ID NO: 6, and an HVR-H3 as set forth in SEQ ID NO: 7. In one

embodiment, the light chain variable region of the antibody comprises an HVR-L1 as set forth in SEQ ID NO: 8, an HVR-L2 as set forth in SEQ ID NO: 9, and an HVR-L3 as set forth in SEQ ID NO: 10. In one embodiment, the antibody comprises a heavy chain variable region as set forth in SEQ ID NO: 3. In one embodiment, the antibody comprises a light chain variable region as set forth in SEQ ID NO: 4. In one embodiment, the antibody comprises a heavy chain variable region as set forth in SEQ ID NO: 3, and a light chain variable region as set forth in SEQ ID NO: 4. In one embodiment, the antibody comprises a k light chain constant region. In one embodiment, the antibody comprises an IgG1 heavy chain constant region.

[0014] In one embodiment, the Fc region of the antibody comprises an amino acid mutation that eliminates fucosylation, such as N297G or N297A. In one embodiment, the Fc region of the antibody comprises an amino acid mutation that enhances binding to FcγRs, such as S239D/1332E or S239D/1332E/A330L. In one embodiment, the antibody comprises a heavy chain as set forth in SEQ ID NO: 12 or 14. In one embodiment, the antibody comprises a light chain as set forth in SEQ ID NO: 11. In one embodiment, the antibody comprises a heavy chain as set forth in SEQ ID NO: 12, and a light chain as set forth in SEQ ID NO: 11. In one embodiment, the antibody comprises a heavy chain as set forth in SEQ ID NO: 14, and a light chain as set forth in SEQ ID NO: 11.

[0015] In one embodiment, the antibody is generated in a host cell deficient in protein fucosylation. In one embodiment, the host cell is α-1,6-fucosyltransferase (FUT8)-deficient, such as gene knockout, decreased expression, and/or decreased activity. In one embodiment, the antibody is generated under conditions where protein fucosylation is inhibited. In one embodiment, the conditions inhibit the expression and/or activity of α-1,6-fucosyltransferase (FUT8). In one embodiment, the host cell is a prokaryotic cell or a eukaryotic cell. In one embodiment, the host cell is *Escherichia coli*. In one embodiment, the cell is a mammalian cell. In one embodiment, the host cell is a CHO cell, such as a CHOK1 cell. In one embodiment, the conditions comprise the addition of an inhibitor of α-1,6-fucosyltransferase (FUT8).

[0016] In another aspect, the present invention provides a composition, comprising the ADCC-enhanced anti-FGFR2 antibody of the present invention. In one embodiment, at least 50%, 60%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or even 100% of the anti-FGFR2 antibodies in the composition are afucosylated.

[0017] In another aspect, the present invention provides a nucleic acid encoding the ADCC-enhanced anti-FGFR2 antibody of the present invention. In another aspect, the present invention provides a vector, comprising the nucleic acid of the present invention. In one embodiment, the vector is a cloning vector or an expression vector. In another aspect, the present invention provides a host cell, comprising the nucleic acid or vector of the present invention.

[0018] In another aspect, the present invention provides a method for generating the ADCC-enhanced anti-FGFR2 antibody of the present invention or the composition comprising the ADCC-enhanced anti-FGFR2 antibody of the present invention. In one embodiment, the method comprises generating the antibody of the present invention in a host cell deficient in protein fucosylation, in order to produce the ADCC-enhanced anti-FGFR2 antibody or compo-

sition of the present invention. In one embodiment, the method comprises culturing the host cell of the present invention under conditions where protein fucosylation is inhibited, in order to produce the ADCC-enhanced anti-FGFR2 antibody or composition of the present invention. In one embodiment, the method comprises culturing the host cell of the present invention under conditions suitable for antibody production, in order to produce the ADCC-enhanced anti-FGFR2 antibody or composition of the present invention, wherein the host cell comprises a nucleic acid encoding an antibody with the fucosylation site deleted. In one embodiment, the host cell is a prokaryotic cell or a eukaryotic cell. In one embodiment, the host cell is *Escherichia coli*. In one embodiment, the cell is a mammalian cell. In one embodiment, the host cell is a CHO cell, such as a CHOK1 cell. In one embodiment, the host cell deficient in protein fucosylation is α-1,6-fucosyltransferase (FUT8)-deficient. In one embodiment, the deficiency of α-1,6-fucosyltransferase (FUT8) is achieved through gene knockout or RNA interference. In one embodiment, the conditions under which protein fucosylation is inhibited are the inhibition of the expression and/or activity of α-1,6-fucosyltransferase (FUT8). In one embodiment, the inhibition of the expression and/or activity of α-1,6-fucosyltransferase (FUT8) is achieved through the addition of an inhibitor of α-1,6-fucosyltransferase (FUT8).

[0019] In another aspect, the present invention provides a pharmaceutical composition, comprising the ADCC-enhanced anti-FGFR2 antibody or composition thereof of the present invention, and at least one pharmaceutically acceptable carrier.

[0020] In another aspect, the present invention provides the ADCC-enhanced anti-FGFR2 antibody, composition, or pharmaceutical composition of the present invention, for use in treating cancer. In another aspect, the present invention provides use of the ADCC-enhanced anti-FGFR2 antibody, composition, or pharmaceutical composition of the present invention for treating cancer. In another aspect, the present invention provides use of the ADCC-enhanced anti-FGFR2 antibody, composition, or pharmaceutical composition of the present invention in the preparation of a medicament for treating cancer. In another aspect, the present invention provides a method for treating cancer, wherein the method comprises administering an effective amount of the ADCC-enhanced anti-FGFR2 antibody, composition, or pharmaceutical composition of the present invention. In one embodiment, the cancer overexpresses FGFR2. In one embodiment, the cancer is gastric cancer or breast cancer. In one embodiment, the medicament is administered in combination with other therapeutic agents.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 shows the antigen binding activity of the antibody of the present invention.

[0022] FIG. 2 shows the ADCC activity of the antibody of the present invention, as detected by using Jurkat as the effector cell. Panel A, KATOIII used as the target cell; Panel B, SUM52PE used as the target cell; and Panel C, SNU16 used as the target cell.

[0023] FIG. 3 shows the ADCC activity of the antibody of the present invention, as detected by using human PBMC as the effector cell. Panel A, KATOIII used as the target cell; Panel B, in SUM52PE used as the target cell; and Panel C, SNU16 used as the target cell.

DETAILED DESCRIPTION OF EMBODIMENTS

[0024] The present invention provides an afucosylated antibody that binds to FGFR2IIIb. In some embodiments, afucosylated antibody heavy and light chains capable of forming an antibody that binds to FGFR2IIIb are also provided. In some embodiments, an afucosylated antibody, heavy chain, and light chain comprising one or more specific hypervariable regions (HVRs) are provided. In some embodiments, compared to a fucosylated anti-FGFR2IIIb antibody, the afucosylated anti-FGFR2IIIb antibody exhibits enhanced ADCC activity. In some embodiments, compared to a fucosylated anti-FGFR2IIIb antibody, the anti-FGFR2IIIb antibody with S239D, 1332E, and A330L mutations in the antibody Fc region exhibits enhanced ADCC activity. In some embodiments, compared to a fucosylated anti-FGFR2IIIb antibody, the afucosylated anti-FGFR2IIIb antibody with S239D, 1332E, and A330L mutations exhibits enhanced ADCC activity.

[0025] A polynucleotide encoding the antibody that binds to FGFR2IIIb is provided. A polynucleotide encoding the antibody heavy or light chain is also provided. A host cell expressing the afucosylated anti-FGFR2IIIb antibody is provided. A method of treatment using the afucosylated anti-FGFR2IIIb antibody with S239D, 1332E, and A330L mutations is provided. Such a method includes, but is not limited to, methods for treating cancers, such as gastric cancer, breast cancer, ovarian cancer, endometrial cancer, pancreatic cancer, and esophageal cancer.

[0026] The present invention provides an antibody targeting FGFR2.

[0027] The present invention also provides a drug combination comprising the antibody of the present invention, and use of the antibody in the preparation of a medicament for diagnosing or treating gastric cancer, especially gastric cancer caused by overexpression of FGFR2IIIb due to gene amplification of FGFR2IIIb.

[0028] Specifically, the present invention encompasses an antibody that binds to FGFR2IIIb and is oligoglycosylated at Asn297, characterized by a fucose content of 75% or more in the oligosaccharide chain. The present invention also provides an ADCC-enhanced anti-FGFR2 antibody, including one in which the fucose at Asn297 of the antibody is knocked out; or S239D/1332E/A330L mutations are introduced at the Fc end of the antibody; or S239D/1332E/A330L mutations are introduced at the Fc end of the afucosylated antibody, which set of antibodies is characterized by enhanced ADCC activity.

[0029] In any of the embodiments described in this application, the antibody may bind to FGFR2IIIb rather than FGFR2IIIC.

[0030] In some embodiments, the antibody is a monoclonal antibody. In some embodiments, the antibody is a chimeric antibody. In some embodiments, the antibody is a humanized antibody. In any of the embodiments described in this application, the antibody may comprise a k light chain constant region. In any of the embodiments described in this application, the antibody may comprise an IgG1 heavy chain constant region.

[0031] In some embodiments, an anti-FGFR2IIIb antibody is provided, wherein the heavy chain variable region comprises: (i) an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 5; (ii) an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 6; and (iii) an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 7; and

the light chain variable region comprises: (iv) an HVR-L1 comprising the amino acid sequence of SEQ ID NO: 8; (v) an HVR-L2 comprising the amino acid sequence of SEQ ID NO: 9; and (vi) an HVR-L3 comprising the amino acid sequence of SEQ ID NO: 10. In some embodiments, the antibody lacks fucose at Asn297. In some embodiments, S239D/1332E/A330L mutations are introduced at the Fc end of the antibody, whose Fc sequence comprises SEQ ID NO: 13. In some embodiments, a composition comprising a plurality of anti-FGFR2IIIb antibodies is provided, wherein the heavy chain CDR region of each anti-FGFR2IIIb antibody in the composition comprises: (i) an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 5; (ii) an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 6; and (iii) an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 7; and the light chain CDR region of each anti-FGFR2IIIb antibody in the composition comprises: (iv) an HVR-L1 comprising the amino acid sequence of SEQ ID NO: 8; (v) an HVR-L2 comprising the amino acid sequence of SEQ ID NO: 9; and (vi) an HVR-L3 comprising the amino acid sequence of SEQ ID NO: 10; and wherein at least 95% of the antibodies in the composition are afucosylated, or S239D/1332E/A330L mutations are introduced at the Fc end of the antibodies in the composition. In some embodiments, the composition may be a supernatant from an antibody-producing cell line. In some embodiments, the composition may be a buffered composition. In some embodiments, the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 3. In some embodiments, the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 4. In some embodiments, the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 3, and the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 4. In some embodiments, the heavy chain comprises the amino acid sequence of SEQ ID NO: 12. In some embodiments, the light chain comprises the amino acid sequence of SEQ ID NO: 11. In some embodiments, the heavy chain comprises the amino acid sequence of SEQ ID NO: 12, and the light chain comprises the amino acid sequence of SEQ ID NO: 11. In some embodiments, the heavy chain comprises the amino acid sequence of SEQ ID NO: 14, and the light chain comprises the amino acid sequence of SEQ ID NO: 11. In some embodiments, a composition comprising a plurality of afucosylated anti-FGFR2IIIb antibodies is provided, wherein the antibodies compete with antibodies comprising the following heavy and light chain variable domains for binding to FGFR2IIIb: the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 3, and the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 4; or the heavy chain of the antibodies comprises the amino acid sequence of SEQ ID NO: 12, and the light chain of the antibodies comprises the amino acid sequence of SEQ ID NO: 11; or the heavy chain of the antibodies comprises the amino acid sequence of SEQ ID NO: 14, and the light chain of the antibodies comprises the amino acid sequence of SEQ ID NO: 11.

[0032] In some embodiments, the afucosylated antibody exhibits enhanced ADCC activity in vitro as compared to a fucosylated anti-FGFR2IIIb antibody with the same amino acid sequence. In some embodiments, the specific lysis caused by the afucosylated anti-FGFR2IIIb antibody is at least 10%, at least 15%, at least 20%, at least 25%, at least

30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 60%, at least 65%, at least 70%, or at least 75% higher than that caused by a fucosylated anti-FGFR2IIIb antibody. In some embodiments, the ADCC activity is determined by using SNU16, KATOIII, and SUM52PE cells expressing FGFR2IIIb as target cells, and isolated human PBMCs as effector cells.

[0033] In some embodiments, the antibody exhibits enhanced ADCC activity in vitro as compared to an anti-FGFR2IIIb antibody with the same amino acid sequence and S239D/1332E/A330L mutations introduced at the Fc end. In some embodiments, the specific lysis caused by the anti-FGFR2IIIb antibody with S239D/1332E/A330L mutations introduced at the Fc end is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 60%, at least 65%, at least 70%, or at least 75% higher than that caused by the original anti-FGFR2IIIb antibody. In some embodiments, the ADCC activity is determined by using SNU16, KATOIII, and SUM52PE cells expressing FGFR2IIIb as target cells, and isolated human PBMCs as effector cells.

[0034] In some embodiments, the afucosylated anti-FGFR2IIIb antibody with S239D/1332E/A330L mutations introduced at the Fc end exhibits enhanced ADCC activity in vitro as compared to an antibody with the same amino acid sequence. In some embodiments, the specific lysis caused by the afucosylated anti-FGFR2IIIb antibody with S239D/1332E/A330L mutations introduced at the Fc end is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 60%, at least 65%, at least 70%, or at least 75% higher than that caused by the original anti-FGFR2IIIb antibody. In some embodiments, the ADCC activity is determined by using SNU16 (a gastric cancer cell line), KATOIII (a gastric cancer cell line), and SUM52PE (a breast cancer cell line) cells expressing FGFR2IIIb as target cells, and isolated human PBMCs as effector cells.

[0035] In some embodiments, a host cell is provided, comprising a nucleic acid encoding the anti-FGFR2IIIb antibody of this application, wherein the host cell lacks the functional α -1,6-fucosyltransferase (FUT8) gene. In some embodiments, the host cell is a CHO cell.

[0036] In some embodiments, a method for manufacturing the afucosylated anti-FGFR2IIIb antibody is provided. In some embodiments, the method comprises culturing the host cell under conditions suitable for expressing a gene encoding the anti-FGFR2IIIb antibody, wherein the host cell lacks the functional α -1,6-fucosyltransferase (FUT8) gene. In some embodiments, the method for manufacturing the afucosylated anti-FGFR2IIIb antibody comprises culturing the host cell under conditions suitable for producing the afucosylated anti-FGFR2IIIb antibody. In some embodiments, the method further comprises recovering the anti-FGFR2IIIb antibody produced by the host cell. In some embodiments, less than 5% of the anti-FGFR2IIIb antibodies produced by the host cell comprise fucose. In some embodiments, at least 95% of the anti-FGFR2IIIb antibodies produced by the host cell lack fucose (i.e., afucosylated). In some embodiments, fucose cannot be detected in the anti-FGFR2IIIb antibodies produced by the host cell. In some embodiments, the presence of fucose is determined by mass spectrometry.

[0037] In some embodiments, a pharmaceutical composition is provided, wherein the pharmaceutical composition

comprises the ADCC-enhanced anti-FGFR2IIIb antibody of this application, and a pharmaceutically acceptable carrier.

[0038] In some embodiments, a method for treating cancer is provided. In some embodiments, the method comprises administering an effective amount of the pharmaceutical composition comprising the ADCC-enhanced anti-FGFR2IIIb antibody of this application, and a pharmaceutically acceptable carrier. In some embodiments, the cancer is selected from gastric cancer, breast cancer, ovarian cancer, endometrial cancer, pancreatic cancer, and esophageal cancer. In some embodiments, the cancer is gastric cancer or breast cancer. In some embodiments, the cancer comprises amplified FGFR2 gene. In some embodiments, the cancer overexpresses FGFR2IIIb. In some embodiments, the cancer comprising overexpressed FGFR2 has a higher degree of overexpression of FGFR2IIIb than FGFR2IIIC. In some embodiments, the cancer overexpresses FGFR2IIIb, but does not comprise amplified FGFR2 gene. In some embodiments, the expression or overexpression of FGFR2IIIb is determined by IHC. In some embodiments, IHC staining of tumor cells as 1+, 2+, or 3+ indicates overexpression of FGFR2IIIb. In some embodiments, IHC staining of tumor cells as 2+ or 3+ indicates overexpression of FGFR2IIIb.

[0039] In some embodiments, the method for treating cancer further comprises administering at least one additional therapeutic agent selected from a platinum agent, paclitaxel, docetaxel, gemcitabine, capecitabine, irinotecan, epirubicin, FOLFOX, FOLFIRI, calcium folinate, fluorouracil, mitomycin C, and doxorubicin hydrochloride in combination. In some embodiments, the platinum agent is selected from cisplatin, oxaliplatin, and carboplatin. In some embodiments, the method for treating cancer further comprises administering paclitaxel. In some embodiments, the method for treating cancer further comprises administering cisplatin and/or 5-FU.

Terminology

[0040] “FGFR2IIIb” or “FGFR2b” are used interchangeably to refer to the splicing form IIIb of fibroblast growth factor receptor 2. A non-limiting exemplary amino acid sequence of mature human FGFR2IIIb is as set forth in SEQ ID NO: 1.

[0041] “FGFR2IIIC” or “FGFR2c” are used interchangeably to refer to the splicing form IIIC of fibroblast growth factor receptor 2. A non-limiting exemplary amino acid sequence of mature human FGFR2IIIC is as set forth in SEQ ID NO: 2.

[0042] The term “antibody” herein is used in the broadest sense, and covers various antibody structures, including, but not limited to, monoclonal antibodies, polyclonal antibodies, multispecific antibodies (e.g., bispecific antibodies), and antibody fragments, so long as they exhibit the desired antigen binding activity.

[0043] The term “antibody” includes, but is not limited to, fragments that are capable of binding to an antigen, such as Fv, single-chain Fv (scFv), Fab, Fab', and (Fab')₂. Papain digestion of antibodies produces two identical antigen-binding fragments, each with a single antigen-binding site, and a residual “Fc” fragment, whose name reflects its ability to crystallize readily. Pepsin treatment yields an F(ab')₂ fragment that has two antigen-binding sites and is still capable of cross-linking an antigen. The term “antibody” also includes, but is not limited to, chimeric antibodies,

humanized antibodies, and antibodies of various species such as mouse, human, macaque, etc.

[0044] The term “heavy chain variable region” refers to a region comprising heavy chain HVR1, framework (FR) 2, HVR2, FR3, and HVR3. In some embodiments, a heavy chain variable region also comprises at least a portion of an FR1 and/or at least a portion of an FR4.

[0045] The term “heavy chain constant region” refers to a region comprising at least three heavy chain constant domains, CH1, CH2, and CH3. Non-limiting exemplary heavy chain constant regions include γ , δ , and α . Non-limiting exemplary heavy chain constant regions also include ϵ and μ . Each heavy chain constant region corresponds to an antibody isotype. For example, an antibody comprising a γ constant region is an IgG antibody, an antibody comprising a δ constant region is an IgD antibody, and an antibody comprising an α constant region is an IgA antibody. Further, an antibody comprising a μ constant region is an IgM antibody, and an antibody comprising an ϵ constant region is an IgE antibody. Certain isotypes can be further subdivided into subclasses. For example, IgG antibodies include, but are not limited to, IgG1 (comprising a $\gamma 1$ constant region), IgG2 (comprising a $\gamma 2$ constant region), IgG3 (comprising a $\gamma 3$ constant region), and IgG4 (comprising a $\gamma 4$ constant region) antibodies; IgA antibodies include, but are not limited to, IgA1 (comprising an $\alpha 1$ constant region) and IgA2 (comprising an $\alpha 2$ constant region) antibodies; and IgM antibodies include, but are not limited to, IgM1 and IgM2.

[0046] The term “heavy chain” refers to a polypeptide comprising at least a heavy chain variable region, with or without a leader sequence. In some embodiments, a heavy chain comprises at least a portion of a heavy chain variable region. The term “full-length heavy chain” refers to a polypeptide comprising a heavy chain variable region and a heavy chain constant region, with or without a leader sequence.

[0047] The term “light chain variable region” refers to a region comprising light chain HVR1, framework (FR) 2, HVR2, FR3, and HVR3. In some embodiments, a light chain variable region also comprises an FR1 and/or an FR4.

[0048] The term “light chain constant region” refers to a region comprising a light chain constant domain, CL. Non-limiting exemplary light chain constant regions include λ and κ .

[0049] The term “light chain” refers to a polypeptide comprising at least a light chain variable region, with or without a leader sequence. In some embodiments, a light chain comprises at least a portion of a light chain constant region. The term “full-length light chain” refers to a polypeptide comprising a light chain variable region and a light chain constant region, with or without a leader sequence.

[0050] The term “hypervariable region” or “HVR” refers to each of the regions of the antibody variable domains which are hypervariable in sequence and/or form structurally defined loops (“hypervariable loops”). Generally, naturally occurring four-chain antibodies comprise six HVRs: three in the VH (H1, H2, and H3), and three in the VL (L1, L2, and L3). The HVR generally comprises amino acid residues from a hypervariable loop and/or from a “complementarity determining region” (CDR), whose sequence has the highest variability and/or which is involved in antigen recognition.

[0051] The term “antibody fragment” comprises a portion of a full-length antibody, generally at least the antigen-binding portion or variable region thereof. Examples of antibody fragments include diabodies, single-chain antibody molecules, conjugates such as immunotoxins, and multispecific antibodies comprising antibody fragments.

[0052] The term “humanized antibody” refers to an antibody in which the framework regions (FRs) and/or “complementarity determining regions” (CDRs) have been modified to comprise the CDR of an immunoglobulin of different specificity as compared to that of the parent immunoglobulin.

[0053] The term “epitope” denotes a protein determinant capable of specifically binding to an antibody. Epitopes usually consist of chemically active surface groupings of molecules such as amino acids or sugar side chains and usually have specific three dimensional structural characteristics, as well as specific charge characteristics. Conformational and nonconformational epitopes are distinguished in that the binding of the antibody to the former but not the latter is lost in the presence of denaturing solvents.

[0054] The term “antibody-dependent cell-mediated cytotoxicity (ADCC)” denotes that killer cells such as NK cells directly kill target cells by recognizing the antibody Fc fragment coated on the target antigen through their expressed Fc receptors.

[0055] The term “fucose” in the present invention refers to fucose at Asn297 in the sugar chain of the antibody. The “fucose content” in the present invention refers to the sum of all fucose residues linked to Asn297, as measured by mass spectrometry and calculated as an average. The “relative amount of fucose” refers to the percentage of the structure containing fucose relative to all sugar structures, as identified by mass spectrometry in samples treated with IdeS.

[0056] The term “afucosylated antibody” is an IgG1 monoclonal antibody with a glycoform core structure dominated by afucosylated glycoforms and a fucose content not exceeding 5%.

[0057] The term “human effector cell” is a leukocyte which expresses one or more FcRs and performs effector functions. In a specific embodiment, the cell expresses at least Fc γ RIII and performs ADCC effector functions. Examples of human leukocytes which can mediate ADCC include peripheral blood mononuclear cells (PBMCs), natural killer (NK) cells, monocytes, macrophages, cytotoxic T cells, and neutrophils. Effector cells can be isolated from natural sources, such as from blood.

[0058] The term “combination therapy” refers to the administration of the anti-FGFR2IIIb antibody of the present invention in combination with chemotherapeutic drugs or immunosuppressants in a single or two separate formulations.

EXAMPLES

Example 1: Design and Preparation of ADCC-Enhanced Anti-FGFR2b Antibody

[0059] According to the amino acid sequences of heavy chain (SEQ ID NO: 12 or 14) and light chain (SEQ ID NO: 11) shown in Table 5 below, gene sequences were synthesized (General Biol, Chuzhou, Anhui), and cloned into the expression vector pcDNATM 3.1 (Invitrogen, V79020), in order to separately prepare plasmids expressing heavy and light chains, which were both separately co-transfected into

CHOK1 and Fut8KO CHOK1 cells according to Table 1 below. The transformed cells were cultured in 20% BM024H+80% BM022H in a shaker at 120 rpm at 37° C. and 6% CO₂ for one week, and the supernatant was collected (WuXi Biologics, Shanghai, China). The supernatant was purified by protein A affinity chromatography column (GE Healthcare). After purification, the purity of the antibodies was detected by the detection methods of SDS-PAGE (Thermo Fisher Scientific, NP0335BOX) and SEC-HPLC (Thermo, UltiMate3000), respectively. The purity of all antibodies was 95% or more.

TABLE 1

Expression of antibody polypeptide chains			
Sample batch No.	Host cell	Heavy chain	Light chain
FWB1914	CHOK 1	SEQ ID NO: 12	SEQ ID NO: 11
FWB19142	CHOK 1	SEQ ID NO: 14	SEQ ID NO: 11
FWB19143	Fut8KO CHOK1	SEQ ID NO: 12	SEQ ID NO: 11
FWB19144	Fut8KO CHOK1	SEQ ID NO: 14	SEQ ID NO: 11

[0060] The antibodies were incubated with IdeS enzyme (Genovis, A0-FR1-020), and Fc/2 and F(ab')₂ fragments were obtained after digestion. The glycoforms were detected by using the ACQUITY UPLC I-Class/Acquity RDa system (Waters), and the relative content of a particular glycoform was analyzed based on the mass spectrometry peak area of the corresponding glycosyl group.

TABLE 2

Analysis results of glycoforms				
	FWB1914	FWB19143	FWB19142	FWB19144
Man5	5.04%	4.28%	1.67%	ND
G0 - GN	ND	3.69%	ND	ND
G0F - GN	8.06%	ND	1.4%	ND
G0	1.81%	88.49%	ND	87.98%
G0F	63.98%	ND	68.69%	ND
G1	13.15%	3.54%	13.91%	12.02%
G1F	7.97%	ND	10.52%	ND
G0F + GN	ND	ND	3.81%	ND

ND: Not detectable

[0061] As shown in Table 2, mass spectrometry analysis showed that the afucosylation in FWB19143 generated in host cells deficient in fucosylation, which had the same sequence as FWB1914, reached 95.77%; and similarly, the afucosylation in FWB19144 generated in host cells deficient in fucosylation, which had the same sequence as FWB19142, reached 100%.

Example 2: Identification of Antigen Binding Activity of Anti-FGFR2b Antibody by ELISA

[0062] A 96-well ELISA plate (Corning, 3590) was first coated with 100 μ L of 2 μ g/mL rhFGFR2b-Fc (Sinobiological, 16485-H02H) overnight at 4° C., and incubated with 250 μ L of blocking solution (PBST containing 3% BSA) for 2 hours at 37° C., and then washed with PBST for 3 times. The antibodies to be tested were subjected to a 5-fold series

dilution from 15 μ g/mL for a total of 8 concentration points (including a blank control), and the serially diluted antibodies were added to the ELISA well at 100 μ L per well. The plate was incubated for 1 hour at 37° C., and then washed with PBST for 3 times. A secondary antibody mouse anti-human Fab-HRP conjugate (Genscript, A01855-200) was added. The plate was incubated for 1 hour at 37° C. Finally, the plate was washed with PBST for 3 times, and 100 μ L of TMB (Biodragon, BF06007) was added for 15 minutes. The chromogenic reaction was terminated by using 2 N sulfuric acid (50 μ L), and the absorbance value at 450 nm was detected on a plate reader (PerkinElmer, EnVision2104) to calculate the binding activity (EC₅₀, nM) of each antibody to human FGFR2b.

[0063] As shown in FIG. 1, the EC₅₀s of antibodies FWB1914, FWB19142, FWB19143, and FWB19144 ranged from 0.57 to 0.71 nM, demonstrating that the modification of the Fc region did not affect the antibody's binding activity to FGFR2b.

Example 3: Detection of ADCC Activity of Antibodies by Using Jurkat Cells as Effector Cells

[0064] KATOIII (a gastric cancer cell line, Chinese Academy of Sciences Cell Bank, SCSP-573), SNU16 (a gastric cancer cell line, Zhongyuan, CRL-5974), or SUM52PE (a breast cancer cell line, BIOIVT, HUMANSUM-0003018) cells with high expression of FGFR2 were used as target cells, respectively. The density of the target cells was adjusted to 0.75 $\times$ 10⁶ cells/mL with phenol red-free RPMI1640 medium containing 10% FBS. The target cells were added to a white 96-well plate (Thermo, 136101) at 100 μ L per well. The plate was cultured in a cell incubator overnight at 37° C. and 5% CO₂. The following day, the culture supernatant was removed, 25 μ L of serum-free, phenol red-free RPMI1640 was added to each well, and 25 μ L of antibodies diluted at a 5-fold concentration gradient starting from a concentration of 1-5 μ g/mL was added, with an equal volume of serum-free, phenol red-free RPMI1640 or negative samples added to the negative wells. The plate was incubated in a cell incubator at 37° C. and 5% CO₂ for 60 minutes. The concentration of Jurkat (Suzhou Rhino Biotechnology, RA-CK01) effector cells was adjusted to 1 $\times$ 10⁶ cells/mL with serum-free, phenol red-free RPMI1640. 25 μ L of cell suspension was added to each well. The plate was incubated in an incubator at 37° C. and 5% CO₂ for 6 hours. The plate to be tested was allowed to return to room temperature, a firefly luciferase detection reagent (Vazyme, DD1204-01) was added, and the plate was incubated at room temperature in dark for 5 minutes, after which the luminescent value was detected.

[0065] As shown in FIG. 2, as detected by using Jurkat cells as effector cells, and KATOIII, SUM52PE, or SNU16 cells with high expression of FGFR2b as target cells, the anti-FGFR2b antibodies, which were afucosylated or had DLE mutations, exhibited stronger ADCC activity than the unmodified wild-type anti-FGFR2b antibody FWB1914. The fold enhancement (calculated based on EC₅₀ (ng/mL)) in ADCC activity of each Fc-modified antibody over the wild-type anti-FGFR2b antibody FWB1914 was shown in Table 3.

TABLE 3

Fold increase in ADCC potency of antibodies with modified Fc region over wild-type antibody			
Target cell	FWB19142	FWB19143	FWB19144
KATOIII	14	12	18
SUM52PE	16	12	22
SNU16	3	3	8

Example 4: Detection of ADCC activity of antibodies by using human PBMCs as effector cells

[0066] KATOIII, SNU16, or SUM52PE cells were used as target cells, respectively. The density of the target cells was adjusted to 0.2×10⁶ cells/mL with phenol red-free RPMI1640 medium containing 10% FBS. The target cells were added to a 96-well plate (Corning, 3599) at 50 μL per well. The plate was cultured in a cell incubator overnight at 37° C. and 5% CO₂. The following day, the culture supernatant was removed, 50 μL of phenol red-free RPMI1640 medium containing 1% FBS was added to each well, and 50 μL of antibodies diluted at a 5-fold concentration gradient starting from a concentration of 1-100 μg/mL was added, with an equal volume of serum-free, phenol red-free RPMI1640 or negative samples added to the negative wells. The plate was incubated in a cell incubator at 37° C. and 5% CO₂ for 60 minutes. Human PBMC (Shanghai Saili Biotechnology, XFBHP025B) effector cells were thawed with phenol red-free RPMI1640 medium containing 10% FBS and IL-2 at a final concentration of 300 IU/mL, and cultured overnight. Then, the cells were centrifuged at 300 g for 10 minutes, resuspended with phenol red-free RPMI1640 con-

taining 1% FBS, and counted. The concentration of the cells was adjusted to 5×10⁶ cells/mL. 50 μL of cell suspension was added to each well. The plate was incubated in a cell incubator at 37° C. and 5% CO₂ for 6 hours. 50 μL of supernatant was sucked out from the well to be tested, 50 μL of LDH detection reagent (Promega, G1781) was added, and the plate was incubated at room temperature in dark for 30 minutes. After adding 50 μL of stop solution, the detection of OD492 was completed within 1 hour.

[0067] As shown in FIG. 3, as detected by using human PBMC cells as effector cells, and KATOIII, SUM52PE, or SNU16 cells with high expression of FGFR2b as target cells, the anti-FGFR2b antibodies, which were afucosylated or had DLE mutations, exhibited stronger ADCC activity than the unmodified wild-type anti-FGFR2b antibody FWB1914. The fold enhancement (calculated based on EC50 (ng/mL)) in ADCC activity of each Fc-modified antibody over the wild-type anti-FGFR2b antibody FWB1914 was shown in Table 4.

TABLE 4

Fold increase in ADCC potency of antibodies with modified Fc region over wild-type antibody			
Target cell	FWB19142	FWB19143	FWB 19144
KATOIII	2.3	1.7	3.4
SUM52PE	2.7	3.8	13.57
SNU16	2.3	1.8	4.3

TABLE 5

Sequences
SEQ ID NO: 1, FGFR2b MVSWGRFICLVVVTMATLSLARPSFSLVEDTTLEPEEPPTKYQISQPEVYVAAPGESLE VRCLLKDAAVISWTKDGVHLGPNNRVTLIGEYLQIKGATPRDSGLYACTASRTVDS EYFMVNVTDALSSGDEDDTDGAEDFVSENSNNKRAPYWTNTEKMEKRLHAVPAA NTVKFRCPAGGNPMPMTMRWLKNGKEFKQEHRIGGYKVRNQHWSLIMESVVPD KGN YTCVVENEYGSINHTYHLDVVERS PHRPILQAGLPANASTVVG GDVEFVCKVYSDAQ PHIQWIKHVEKNGSKYGPDGLPYLKV LKHSGINSNAEVLALFNVTADAGEYICKVS NYIGQANQSAWLT VLPKQQA GREKEITASPDYLEIAIYCIGVFLIACMVVTVILCRM KNTTKPDFSQPAVHKLTKRIPLRQVTVSAESSSSMNSNTPLVRIT TRLSSADT PML AGVSEYELPEDPKWEPPRDKLTGKPLGEGCF GQVMAEAVGIDKDKPKEAVTVAV KMLKDDATEKDLSDLVSE MEMMKMIGKHKNIIINLLGACTQDGPPLYVIEYASKGNLR EYLRARRPPGMEYSYDINRVPEEQMTFKDLVSCTYQLARGMEY LASQKCIHRDLAARNV LVLTENNVMKIADFG LARDINNIDYKKT TNGRLPVKWM APEALFDRVYTHQSDV WSPGVLMWEIFTLGGS PYPGIPVEELFKLLKEGHRMDKPANCTNELYMMMRDCWHA VPSQRPTFKQLVEDLDRILTLTTNEEYLDLSQPLEQYSPSPD TRSSCSSGDDSVFSPDPMPEYEPCLPQYPHINGSVKT
SEQ ID NO: 2, FGFR2c MVSWGRFICLVVVTMATLSLARPSFSLVEDTTLEPEEPPTKYQISQPEVYVAAPGESLE VRCLLKDAAVISWTKDGVHLGPNNRVTLIGEYLQIKGATPRDSGLYACTASRTVDS EYFMVNVTDALSSGDEDDTDGAEDFVSENSNNKRAPYWTNTEKMEKRLHAVPAA NTVKFRCPAGGNPMPMTMRWLKNGKEFKQEHRIGGYKVRNQHWSLIMESVVPD KGN YTCVVENEYGSINHTYHLDVVERS PHRPILQAGLPANASTVVG GDVEFVCKVYSDAQ PHIQWIKHVEKNGSKYGPDGLPYLKV LKAAGVNTDKEIEVLVIRNVT FEDAGEYTCL AGNSIGISFHS AWLTVLPAPGREKEITASPDYLEIAIYCIGVFLIACMVVTVILCRM KNTTKKPDFSQPAVHKLTKRIPLRQVTVSAESSSSMNSNTPLVRIT TRLSSADT PMLAG VSEYELPEDPKWEPPRDKLTGKPLGEGCF GQVMAEAVGIDKDKPKEAVTVAVKM LKDDATEKDLSDLVSE MEMMKMIGKHKNIIINLLGACTQDGPPLYVIEYASKGNLR EYLRARRPPGMEYSYDINRVPEEQMTFKDLVSCTYQLARGMEY LASQKCIHRDLAARNV LVLTENNVMKIADFG LARDINNIDYKKT TNGRLPVKWM APEALFDRVYTHQSDVWS FGVLMWEIFTLGGS PYPGIPVEELFKLLKEGHRMDKPANCTNELYMMMRDCWHA VP SQRPTFKQLVEDLDRILTLTTNEEYLDLSQPLEQYSPSPD TRSSCSSGDDSVFSPDPMPEYEPCLPQYPHINGSVKT

Sequences

SEQ ID NO: 14, heavy chain (DLE mutations)
(variable domain is single-underlined, and Fc region is
double-underlined)

QVQLVQSGAEVKKPGSSVKVSKASGYIFTSYNNVHWVRQAPGQGLEWIGSIYPDNGD
SSYNNNYKGRATITADKSTSTAYMELSSLRSEDAVVYVCARGDFAYWGQGLTVTVSS
ASTKGPSVFPLAPSSKSTSGTAALGCLVKVDYFPEPVTVSWNSGKSLTSGVTHFPFAPVLQ
SSGLYSLSVSVFPPSSSLGTQTYICNVNKKPSNTKVDKRVESPKCDKTHCTCPCPAPEL
LGDDPFLFLPPKPKDITLMSRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAAKTPK
REEQYNSTYRVVSVLTVLHQDWLNGKEYCKVCKVSNKALPPEKTTISKAKGQPREPQV
YTLPPSREEMTKNQVSLTCLVGLFQPSDIAVEENSGLPENNYKTTTPVLDSGDSFEL
YSLTIVDKSRWQGNVFSQVMHEALHNHYTOKSLSLSPGK

SEQUENCE LISTING

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Sequence total quantity: 14
SEQ ID NO: 1          moltype = AA length = 822
FEATURE               Location/Qualifiers
source                1..822
                     mol_type = protein
                     note = FGFR2b
                     organism = synthetic construct
```

-continued

SEQUENCE: 1

MVSWGRFICL	VVVTMATLSL	ARPSFSLVED	TITLEPEEPPT	KYQISQPEVY	VAAPGESLEV	60
RCLLKDAAVI	SWTKDGVHLG	PNNRTVLIGE	YLQIKGATPR	DSGLYACTAS	RTVDSETWYF	120
MVNVTDIAIS	GDDEDDTDGA	EDFVSENSNN	KRAPYWTNTE	KMEKRLHAVP	AANTVKFRCP	180
AGGNPMPTMR	WLKNGKEFKQ	EHRIGGYKVR	NQHWSLIMES	VVPSDKGNYT	CVVENEYSGI	240
NHTYHLDVVE	RSPHRPILQA	GLPANASTVV	GGDVEFVCKV	YSDAQPHIQW	IKHVEKNGSK	300
YGPDGLPYLK	VLKHSGINSS	NAEVLALFNV	TEADAGEYIC	KVSNYIGQAN	QSAWLTVLPK	360
QQAPGREKEI	TASPDYLEIA	IYCIGVFLIA	CMVVTVILCR	MKNNTKKPDF	SSQPAVHKLT	420
KRIPLRRQVT	VSAESSSSMN	SNTPLVIRIT	RLSSTADTPM	LAGVSEYELP	EDPKWEFPRD	480
KLTLGKPLGE	GCFGQVVMAE	AVGIDKDKPK	EAVTVAVKML	KDDATEKDLS	DLVSEMEMMK	540
MIGKHKNIIN	LLGACTQDGP	LYVIVEYASK	GNLREYLRRR	RPPGMEYSYD	INRVPEEQMT	600
FKDLVSTCTY	LARGMEYLAS	QKCIHRDLAA	RNVLVTEENV	MKIADFGLAR	DINNIDYKKG	660
TTNGRLPVKM	MAPEALFDRV	YTHQSDVWSF	GVLMEIIFTL	GGSPYPGIPV	EELFKLLKEG	720
HRMDKPANCT	NELYMMMRDC	WHAVPSQRPT	FKQLVEDLDR	ILTLTTNEEY	LDLSQPLEQY	780
SPSYPDTRSS	CSSGDDSVFS	PDMPYEPCL	PQYPHINGSV	KT		822

SEQ ID NO: 2 moltype = AA length = 821
 FEATURE Location/Qualifiers
 source 1..821
 mol_type = protein
 note = FGFR2c
 organism = synthetic construct

SEQUENCE: 2

MVSWGRFICL	VVVTMATLSL	ARPSFSLVED	TITLEPEEPPT	KYQISQPEVY	VAAPGESLEV	60
RCLLKDAAVI	SWTKDGVHLG	PNNRTVLIGE	YLQIKGATPR	DSGLYACTAS	RTVDSETWYF	120
MVNVTDIAIS	GDDEDDTDGA	EDFVSENSNN	KRAPYWTNTE	KMEKRLHAVP	AANTVKFRCP	180
AGGNPMPTMR	WLKNGKEFKQ	EHRIGGYKVR	NQHWSLIMES	VVPSDKGNYT	CVVENEYSGI	240
NHTYHLDVVE	RSPHRPILQA	GLPANASTVV	GGDVEFVCKV	YSDAQPHIQW	IKHVEKNGSK	300
YGPDGLPYLK	VLKAAVNTT	DKEIEVLVIR	NVTFEDAGEY	TCLAGNSIGI	SPHSAWLTVL	360
PAPGREKEIT	ASPDYLEIAI	YCIGVFLIAC	MVVTVILCRM	KNTTKKPDFS	SQPAVHKLTK	420
RIPLRQVTV	SAESSSSMNS	NTPLVRITTR	LSSTADTPML	AGVSEYELPE	DPKWEFPRDK	480
LTLGKPLGEG	CFGQVVMAEA	VGIDKDKPKE	AVTVAVKMLK	DDATEKDLS	LVSEMEMMKM	540
IGKHKNIINL	LGACTQDGPL	YVIVEYASKG	NLREYLRRRR	PPGMEYSYDI	NRVPEEQMTF	600
KDLVSTCTYQ	ARGMEYLASQ	KCIHRDLAAR	NVLVTENNVM	KIADFGGLAR	INNIDYKKT	660
TNGRLPVKWM	APEALFDRVY	THQSDVWSFG	VLMWEIIFTLG	GSPYPGIPVE	ELFKLLKEGH	720
RMDKPANCTN	ELYMMMRDCW	HAVPSQRPTF	KQLVEDLDRI	LTLTTNEEYL	DLSQPLEQYS	780
PSYPDTRSSC	SSGDDSVFSP	DPMPYEPCLP	QYPHINGSVK	T		821

SEQ ID NO: 3 moltype = AA length = 114
 FEATURE Location/Qualifiers
 source 1..114
 mol_type = protein
 note = VH
 organism = synthetic construct

SEQUENCE: 3

QVQLVQSGAE	VKKPGSSVKV	SCKASGYIFT	SYNVHWRQA	PGQGLEWIGS	IYPDNGDSSY	60
NNNYKGRATI	TADKSTSTAY	MELSSLRSED	TAVYYCARGD	FAYWGQGLTV	TVSS	114

SEQ ID NO: 4 moltype = AA length = 107
 FEATURE Location/Qualifiers
 source 1..107
 mol_type = protein
 note = VL
 organism = synthetic construct

SEQUENCE: 4

DIQMTQSPSS	LSASVGRVDT	ITCKASNGVS	NDIAWYQQKP	GKAPKLLIYS	ASYRYSGVPS	60
RFSGSGSGTD	FTFTISLQP	EDIATYYCQ	HSTTPYTFGQ	GTKLEIK		107

SEQ ID NO: 5 moltype = AA length = 5
 FEATURE Location/Qualifiers
 source 1..5
 mol_type = protein
 note = HVR-H1
 organism = synthetic construct

SEQUENCE: 5

SYNVH		5
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SEQ ID NO: 6 moltype = AA length = 17
 FEATURE Location/Qualifiers
 source 1..17
 mol_type = protein
 note = HVR-H2
 organism = synthetic construct

SEQUENCE: 6

SIYPDNGDSS	YNNNYKG	17
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-continued

SEQ ID NO: 7	moltype = AA length = 5	
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source	1..5	
	mol_type = protein	
	note = HVR-H3	
	organism = synthetic construct	
SEQUENCE: 7		
GDFAY		5
SEQ ID NO: 8	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	note = HVR-L1	
	organism = synthetic construct	
SEQUENCE: 8		
KASNGVSNDI A		11
SEQ ID NO: 9	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	note = HVR-L2	
	organism = synthetic construct	
SEQUENCE: 9		
SASYRYS		7
SEQ ID NO: 10	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	note = HVR-L3	
	organism = synthetic construct	
SEQUENCE: 10		
QQHSTTPYT		9
SEQ ID NO: 11	moltype = AA length = 214	
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source	1..214	
	mol_type = protein	
	note = light chain	
	organism = synthetic construct	
SEQUENCE: 11		
DIQMTQSPSS LSASVGDVRT ITCKASNGVS NDIANYQQKP GKAPKLLIYS ASYRYSGVPS		60
RFSGSGSGTD FTFTISLQP EDIATYYCQQ HSTTPYTFGQ GTKLEIKRTV AAPSVFIFPP		120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT		180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC		214
SEQ ID NO: 12	moltype = AA length = 444	
FEATURE	Location/Qualifiers	
source	1..444	
	mol_type = protein	
	note = heavy chain	
	organism = synthetic construct	
SEQUENCE: 12		
QVQLVQSGAE VKKPGSSVKV SCKASGYIFT SYNVHWVRQA PGQGLEWIGS IYPDNGDSSY		60
NNNYKGRATI TADKSTSTAY MELSSLRSED TAVYYCARGD PAYWGQGTLV TVSSASTKGP		120
SVFPLAPSSK STSGGTAALG CLVKDYFPEP VTVSWNSGAL TSGVHTFPAV LQSSGLYSLS		180
SVVTVPSSSL GTQTYICNVN HKPSNTKVDK RVEPKSCDKT HTCPPCPAPE LLGGPSVFLF		240
PPKPKDTLMI SRTPEVTCVV VDVSHEDPEV KFNWYVDGVE VHNAKTKPRE EQYNSTYRVV		300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV		360
SLTCLVKGFY PSDIAVEWES NGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWQQGNVF		420
SCSVMHEALH NHYTQKSLSL SPGK		444
SEQ ID NO: 13	moltype = AA length = 231	
FEATURE	Location/Qualifiers	
source	1..231	
	mol_type = protein	
	note = Fc (DLE mutation)	
	organism = synthetic construct	
SEQUENCE: 13		
PKSCDKTHTC PPCAPELLG GPDVFLFPPK PKDTLMISRT PEVTCVVVDV SHEDPEVKFN		60
WVVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPLPEEKTI		120
SKAKGQPREP QVYTLPPSRE EMTKNQVSLT CLVKGFPSPD IAVEWESNGQ PENNYKTPP		180
VLDSDGSFFL YSKLTVDKSR WQQGNVFSK VMHEALHNHY TQKSLSLSPG K		231

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SEQ ID NO: 14      moltype = AA  length = 444
FEATURE            Location/Qualifiers
source              1..444
                    mol_type = protein
                    note = heavy chain [DLE mutation]
                    organism = synthetic construct

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SEQUENCE: 14
QVQLVQSGAE VKKPGSSVKV SCKASGYIFT SYNHWVRQA PGQGLEWIGS IYPDNGDSSY 60
NNNYKGRATI TADKSTSTAY MELSSLRSED TAVYYCARGD FAYWGQGLTV TVSSASTKGP 120
SVFPLAPSSK STSGGTAALG CLVKDYFPEP VTSWNSGAL TSGVHTFPAPV LQSSGLYSLS 180
SVVTVPSSSL GTQTYICNVN HKPSNTKVDK RVEPKSCDKT HTPPCCPAPE LLGGPDVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVSHEDPEV KFNWYVDGVE VHNAKTKPRE EQYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPLPEE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCLVKGFY PSDIAVEWES NGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWQQGNVF 420
SCSVMEALH NHYTQKSLSL SPGK                                     444

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1. An ADCC-enhanced anti-FGFR2 antibody, comprising a heavy chain variable region, a light chain variable region, and an Fc region,

wherein the heavy chain variable region comprises:

- (i) an HVR-H1 consisting of the amino acid sequence of SEQ ID NO: 5;
- (ii) an HVR-H2 consisting of the amino acid sequence of SEQ ID NO: 6; and
- (iii) an HVR-H3 consisting of the amino acid sequence of SEQ ID NO: 7,

wherein the light chain variable region comprises:

- (iv) an HVR-L1 consisting of the amino acid sequence of SEQ ID NO: 8;
- (v) an HVR-L2 consisting of the amino acid sequence of SEQ ID NO: 9; and
- (vi) an HVR-L3 consisting of the amino acid sequence of SEQ ID NO: 10,

and

wherein compared to a wild-type Fc region that is fucosylated, the Fc region of the antibody exhibits enhanced binding to Fc-gamma receptors (FcγRs).

2. The antibody according to claim 1, wherein the heavy chain variable region consists of the amino acid sequence of SEQ ID NO: 3, or the light chain variable region consists of the amino acid sequence of SEQ ID NO: 4.

3. The antibody according to claim 1, wherein the antibody comprises a k light chain constant region and/or an IgG1 heavy chain constant region.

4. The antibody according to claim 1, wherein the heavy chain of the antibody consists of the amino acid sequence of SEQ ID NO: 12, and the light chain of the antibody consists of the amino acid sequence of SEQ ID NO: 11.

5. The antibody according to claim 1, wherein the Fc region comprises a mutation that enhances binding to FcγRs.

6. The antibody according to claim 1, wherein the heavy chain of the antibody consists of the amino acid sequence of SEQ ID NO: 14, and the light chain of the antibody consists of the amino acid sequence of SEQ ID NO: 11.

7. The antibody according to claim 1, wherein the Fc region is afucosylated.

8. The antibody according to claim 7, wherein the antibody is generated in a host cell deficient in protein fucosylation.

9. The antibody according to claim 8, wherein the host cell is α -1,6-fucosyltransferase (FUT8)-deficient.

10. The antibody according to claim 7, wherein the antibody is generated under conditions where protein fucosylation is inhibited.

11. The antibody according to claim 10, wherein the conditions inhibit α -1,6-fucosyltransferase (FUT8).

12. The antibody according to claim 7, wherein the Fc region comprises a mutation that eliminates fucosylation at N297.

13. A composition; comprising the antibody according to claim 1.

14. The composition according to claim 13, wherein at least 95% of the anti-FGFR2 antibodies in the composition are afucosylated.

15. A host cell, comprising a nucleic acid encoding the antibody according to claim 1.

16. The host cell according to claim 15, wherein the host cell is deficient in protein fucosylation.

17. The host cell according to claim 16, wherein the host cell is FUT8-deficient.

18. A host cell, comprising a nucleic acid encoding the antibody according to claim 12.

19. A method for generating an ADCC-enhanced anti-FGFR2 antibody, the method comprising culturing the host cell according to claim 15 under conditions where protein fucosylation is inhibited.

20. A method for generating an ADCC-enhanced anti-FGFR2 antibody, the method comprising culturing the host cell according to claim 16 under conditions suitable for antibody production.

21. A method for generating an ADCC-enhanced anti-FGFR2 antibody, the method comprising culturing the host cell according to claim 18 under conditions suitable for antibody production.

22. A pharmaceutical composition; comprising the antibody according to claim 11, and a pharmaceutically acceptable carrier.

23. A method of treating cancer in an individual, the method comprising administering a therapeutically effective amount of the antibody according to claim 11 to the individual.

24. The method according to claim 23, wherein the cancer overexpresses FGFR2.

25. The method according to claim 23, wherein the cancer is gastric cancer or breast cancer.

26-28. (canceled)

29. The antibody according to claim 5, wherein the mutation that enhances binding to FcγRs comprises the S239D/1332E mutations or S239D/1332E/A330L mutations according to Kabat's EU index.

30. The antibody according to claim **12**, wherein the mutation that eliminates fucosylation at N297 comprises the N297G mutation or the N297A mutation according to Kabat's EU index.

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